k-mer and GWAS approaches to identify host-specific genomic determinants in *Klebsiella pneumoniae*

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Highlights

- Combination of k-mer count genomic representations with Supervised Machine Learning (ML) models results in high accuracy of *K. pneumoniae* host prediction
- Validation of previous findings and report of novel enriched homolog gene clusters through microbial GWAS
- Automated *in silico* workflow with potential application in epidemiological surveillance and risk assessment of inter-species bacterial pathogen transmission

Introduction

- *K. pneumoniae* is an important Gram-negative opportunistic bacterial pathogen now recognized by the World Health Organization as an urgent threat to public health worldwide (1).
- The epidemiology of *K. pneumoniae* has not been extensively studied and reservoirs of the organism have been rarely investigated (2).
- In this study, we developed a computational workflow aiming to classify *K. pneumoniae* isolates based on their host-of-origin.
- Pangenome and GWAS analyses were further employed for identification and characterization of potential host-associated genetic signatures.

Dataset

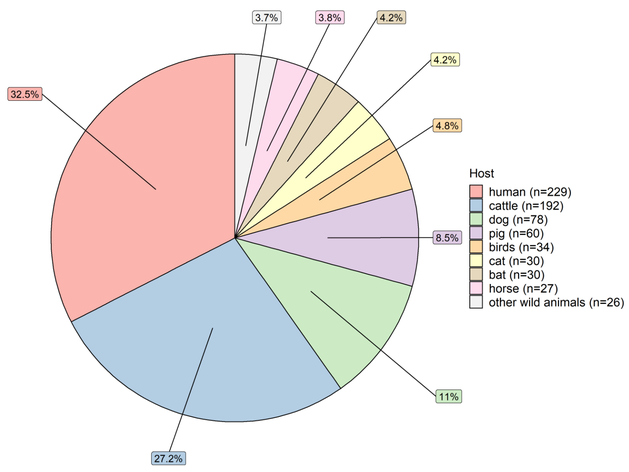

Fig 1. 706 *K. pneumoniae* (NCBI Taxonomy Identifier: 573) isolates used for the analysis, following curation and filtering. Whole-genome sequencing (WGS) data were collected from European Nucleotide Archive (ENA) Pathogens database on October 2021.

K-mer counting, data pre-processing and classification model training

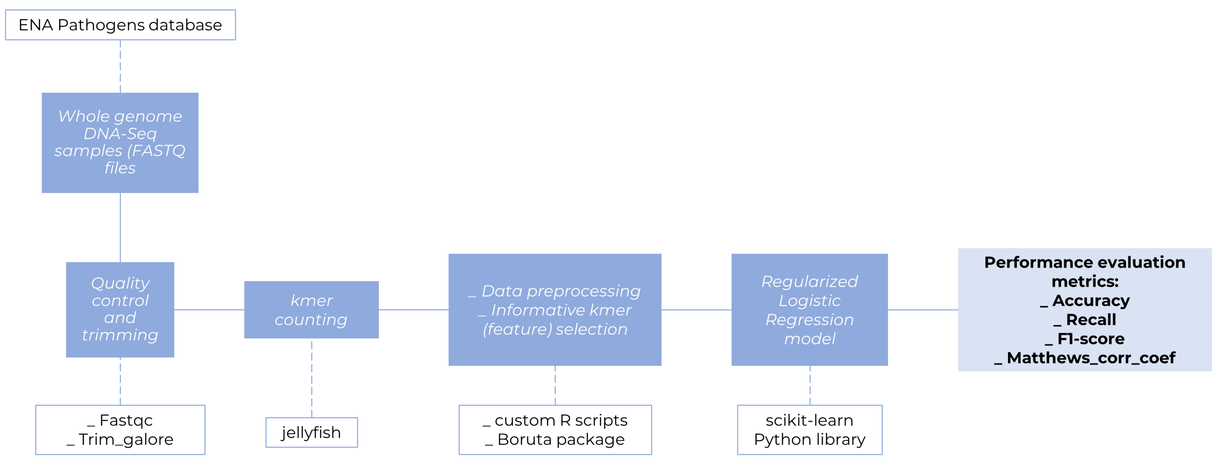

Fig 2. Schematic display of the k-mer count representations and logistic regression model classification workflow. Regularized logistic regression models were trained for host prediction on genomic, k-mer count representations.

Pangenome and GWAS analyses

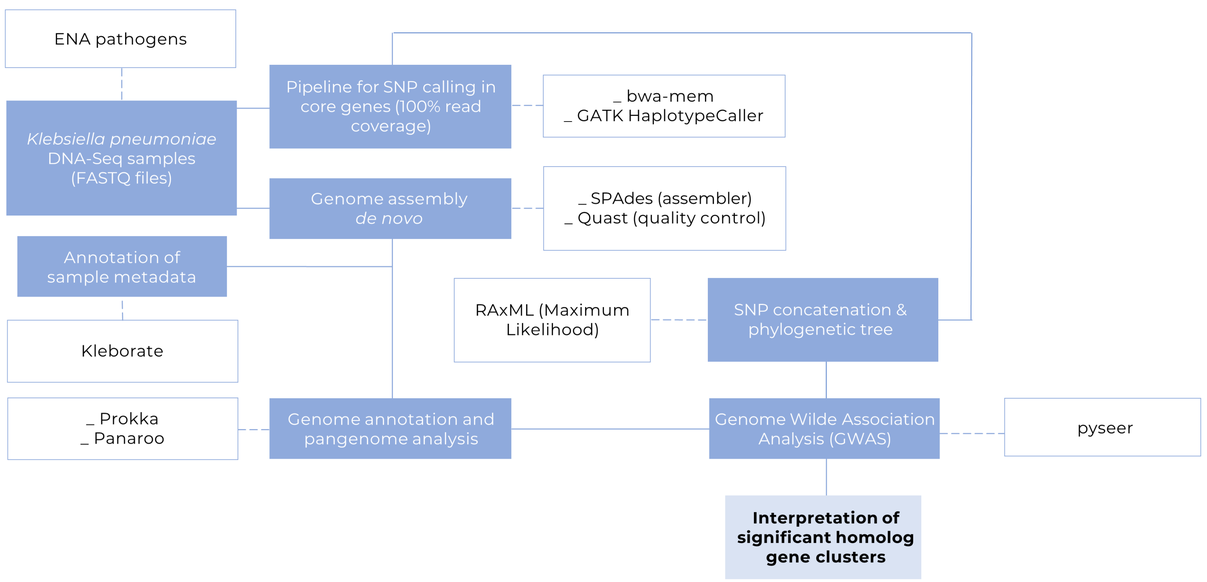

Fig 3. Schematic representation of the microbial pangenome and genome-wide association (GWAS) analyses workflow.

Acknowledgements

This research has been co-financed by the European Union and Greek national funds through the Operational Program Competitiveness, Entrepreneurship and Innovation, under the call RESEARCH – CREATE – INNOVATE (GenOptics, project code: T1EDK-00407).

Dimension reduction of k-mer count genomic representations

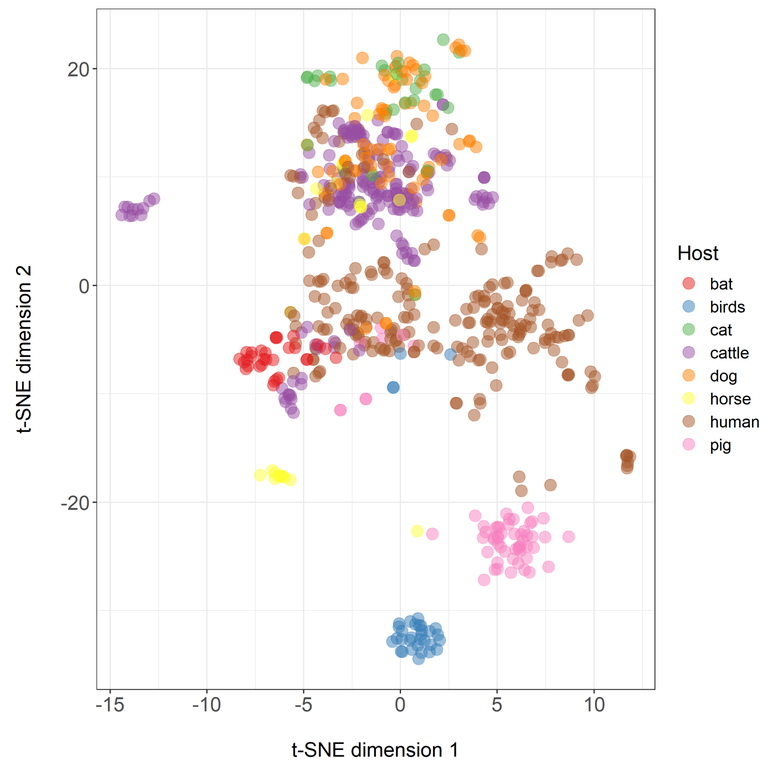

Fig 4. Dimension reduction analysis of k-mer count genomic representations using the t-Distributed Stochastic Neighbor Embedding (t-SNE) algorithm. Dimension reduction analysis was applied to 680 *K. pneumoniae* isolates, following k-mer selection with Boruta algorithm (5) and strong-correlation pruning.

Predictive performance of Regularized Logistic Regression models

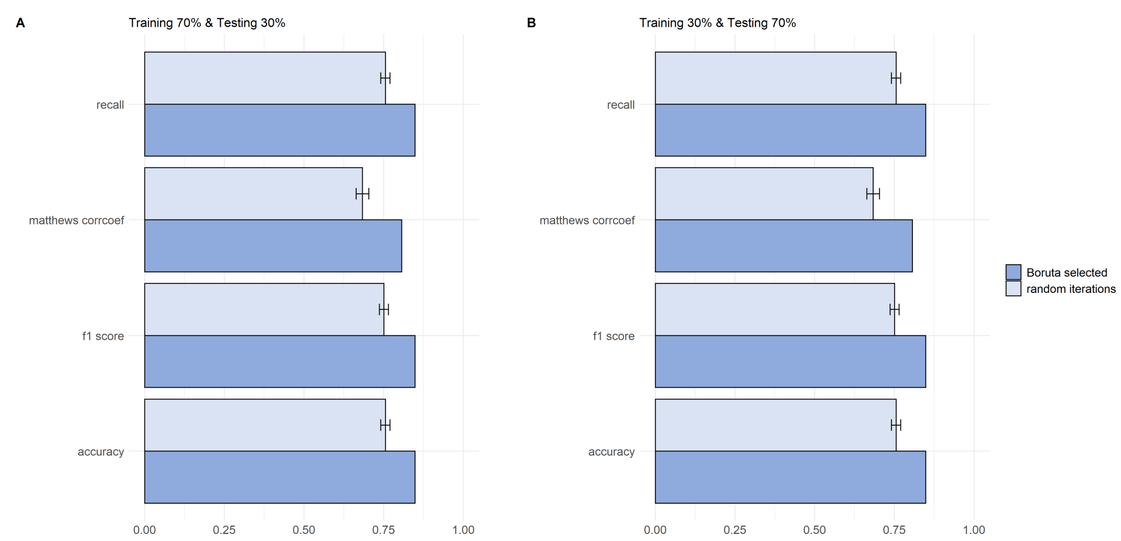

Fig 5. Comparisons of model performance for logistic regression models trained on k-mer count representations of Boruta-selected and correlation-pruned k-mers ("Boruta-selected") against 10 subsets of randomly selected k-mers ("random iterations"). For random iterations, the average value of each metric was used, with error bars corresponding to the standard deviation values. **A**. First set of classification models trained with 70% (training set) and evaluated on 30% (testing set). **B**. Second set of classification models trained with 30% (training set) and evaluated on 70% (testing set).

Host-enriched homolog gene clusters of *K. pneumoniae* isolates

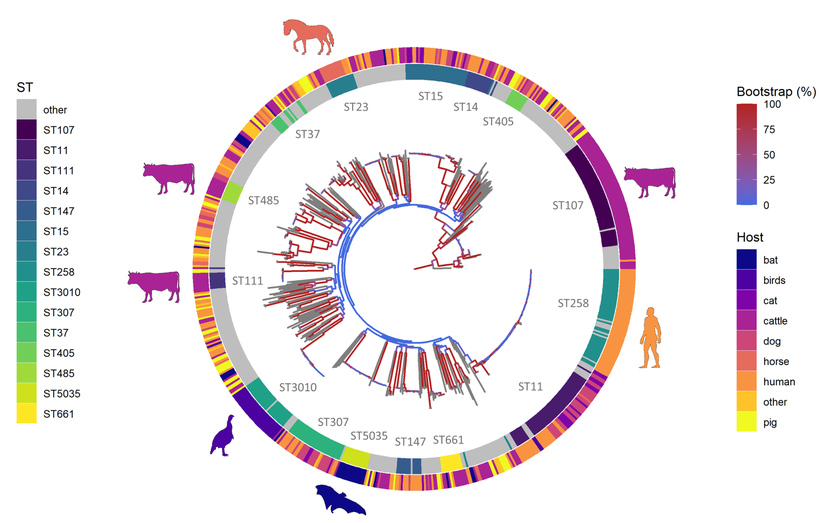

Fig 6. Phylogenetic tree of concatenated core gene SNPs from 706 *K. pneumoniae* isolates (see Methods). Two samples (SAMN18104131, SAMN06437927) were excluded for visualization purposes. Support values (%) for Maximum Likelihood analysis, calculated after 500 bootstrap iterations, are shown with a red-blue color scale.

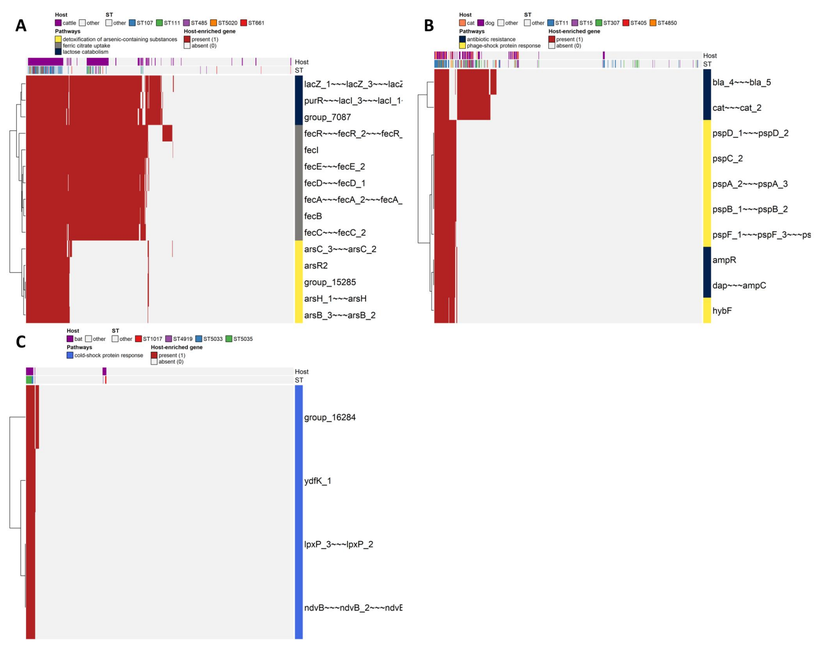

Fig 7. Heatmaps displaying selected homolog gene clusters that were found to be enriched (Bonferroni adjusted p.value < 1e-05) in *K. pneumoniae* isolates of (**A**) cattle, (**B**) companion animals (dog & cat), and (**C**) bat. Heatmaps are annotated for the pathways in which the enriched gene clusters participate (rows), as well as the host categories and the most represented STs in host isolates (columns).